Two moving orbits in a singleton prefix

Human Compression project: L044 continuation

Retain the critical singleton setting with $n = 2r \geq 14$, actual natural $G_0 = \mathrm{Alt}(\Omega \setminus \{a\})$, full ambient IP and the acquired strict prefix-growth proof. For $4 \leq g < r$ write

$$P = G_{<g}, \quad Y = \mathrm{supp}(P), \quad C = a^P, \quad H = G_{\{2,\dots,g-1\}}, \quad b = a^{\rho_0}.$$

Assume P is intransitive on Y and $|Y| \leq 2g + 1$. In particular Y is proper in Ω, because $|Y| \leq 2r - 1 < n$. The points a, b are fixed by H.

Theorem 1. *The group H cannot have precisely two nontrivial orbits and precisely the two fixed points a, b within Y. This exclusion does not require a paired-edge hypothesis.*

Proof. Suppose the stated orbit decomposition holds, and name the nontrivial orbits U, V. Strict prefix growth gives $|C| \geq g + 1 \geq 5$. Thus C meets at least one of U, V, and contains every H-orbit it meets. Since $a, b \in C$, meeting both would make P transitive on Y. Relabel so that

$$C = \{a, b\} \dot\cup V, \qquad E = U. \tag{1}$$

These are the two P-orbits in Y, so ρ_1 preserves both sets.

The mark ρ_0 centralizes H, and therefore permutes its nontrivial orbits equivariantly. If it exchanges U, V, then it is fixed-point-free on their union; if it preserves one of them, its restriction is either the identity or a free involution. This last assertion follows because a centralizing permutation with a fixed point fixes a whole transitive orbit.

If ρ_0 exchanges U, V, or preserves both and is free on both, then $\mathrm{supp}(\rho_0) = Y$. The mark ρ_1 preserves Y, and every mark of index at least two centralizes ρ_0 and preserves its support. Thus Y is a proper nonempty G-invariant set, a contradiction. If ρ_0 is the identity on both U, V, it is just $(a\ b)$ and is odd, again impossible. Consequently it is free on exactly one of the two orbits and fixes the other pointwise.

If it is free on V and fixes E, then $\mathrm{supp}(\rho_0) = C$ by (1). The mark ρ_1 preserves C, and every later mark centralizes ρ_0. Thus C is a proper G-invariant set, a contradiction. The sole remaining case is

$$\rho_0|_E \text{ is free}, \qquad \rho_0|_V = 1. \tag{2}$$

Put $P_1 = \langle \rho_0, \rho_1 \rangle$. Strict growth makes $x = b^{\rho_1} \neq b$. Since ρ_1 fixes a and preserves the two sets in (1), $x \in V$. The P_1-orbit of a is exactly $\{a, b, x\}$: on it the marks are $(a\ b)$ and $(b\ x)$. Actual natural G_0 and full IP give

$$(P_1)_a = P_1 \cap G_0 = \langle \rho_1 \rangle.$$

Orbit–stabilizer therefore gives $|P_1| = 3 \cdot 2 = 6$. In particular $P_1 \cong S_3$ and $(\rho_0\rho_1)^3 = 1$ in the actual group, including its action outside $\{a, b, x\}$.

The set $V \setminus \{x\}$ is preserved by both marks: ρ_0 fixes it, and ρ_1 preserves C and has already paired b with x and fixed a. On this set $\rho_0\rho_1 = \rho_1$ has order dividing two. Its order also divides three by the displayed relation, so both marks fix $V \setminus \{x\}$ pointwise. Every point outside Y is fixed as well.

Every P_1-orbit in E has size dividing six. It cannot have size one or three, because ρ_0 is free there and an involution on an odd set has a fixed point. Hence these orbits have size two or six. We have proved that E is exactly the union of all P_1-orbits of sizes two and six in the whole ambient set; the sole other nontrivial orbit is $\{a, b, x\}$ of size three. Every mark of index at least three centralizes P_1, so preserves this union E. The first two marks preserve it by (1), and $\rho_2 \in H$ preserves it as an H-orbit. Thus every mark preserves E, the final contradiction. □

Consequences and limits. In the numerical budget for a small singleton prefix, this theorem eliminates the entire row $f = 2, c = 2$, including both possible support sizes $2g$ and $2g + 1$. The remaining rows $f = 3, c = 2$ and $f = 2, c = 3$, both with support $2g + 1$, require separate arguments. The theorem above assumes its literal orbit decomposition directly; it does not depend on the status of the separate budget packet.

Attribution. The actual singleton point stabilizer, strict prefix growth and critical range are acquired. The support-invariance split and the use of the actual order-six first prefix to distinguish all exterior orbits are this continuation. No faithfulness of a general orbit projection is used. The source context is Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for A_n*, Propositions 5.28–5.32. No finite census, formal declaration, whole-original recovery or question credit is added.